

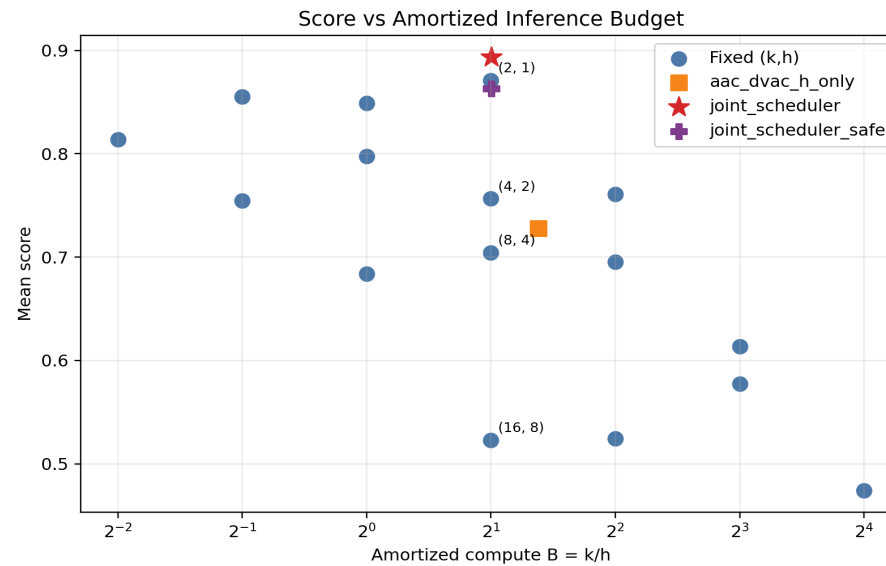
Joint Compute-Control Budgeting

Iso-compute denoising and replanning for Diffusion Policy-style control

- Final project: reproduce a robotics paper and add an improvement idea
- Base paper: Diffusion Policy, RSS 2023
- Task: Push-T-style closed-loop contact benchmark

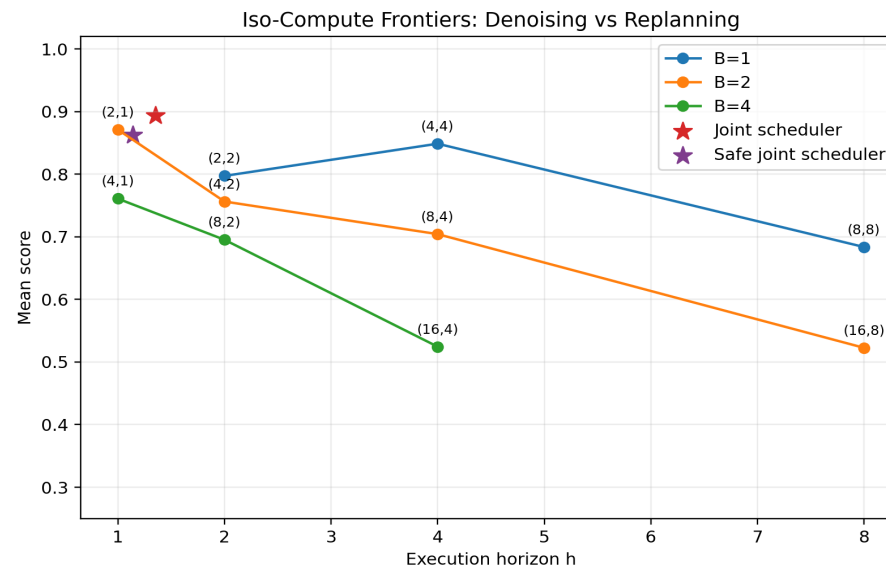
Motivation

- Diffusion Policy is accurate but iterative denoising is expensive
- Replanning improves reactivity but increases policy calls
- Deployment needs a fixed per-step inference budget



Key Question

- Should compute be spent on larger k or smaller h?
- k: denoising steps per policy call
- h: executed controls before replanning
- $B \approx k / h$ is the iso-compute budget

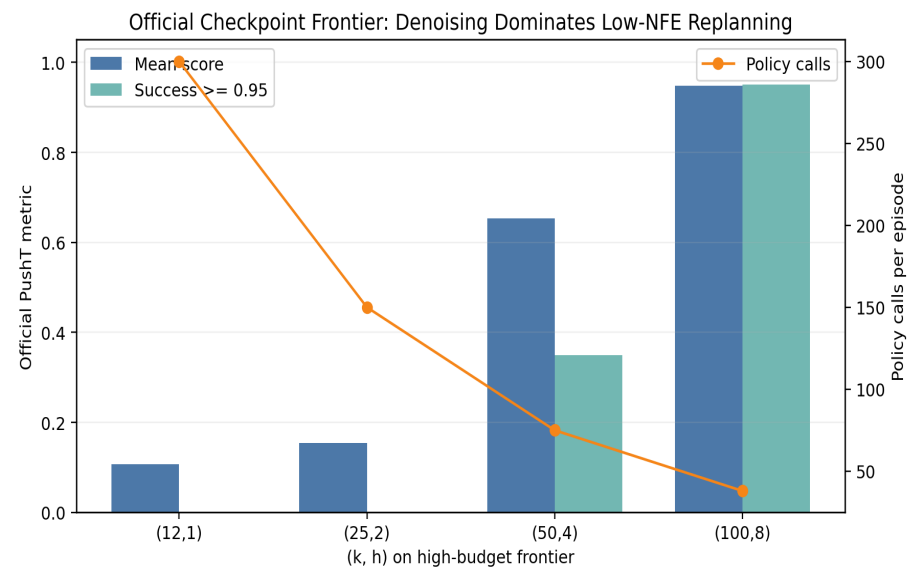


Reproduction Setup

- Official source acquired via GitHub codeload archive
- Official Push-T checkpoint downloaded: 1044185793 bytes
- 50-seed official runner score: 0.919
- Compatibility patches needed for Python 3.12 / Gym 0.26

Official Frontier

- 20-seed official fixed-(k,h) sweep
- Best tested official point: (100,8)
- Score: 0.949, success: 0.950
- Low-k frequent replanning does not rescue this checkpoint

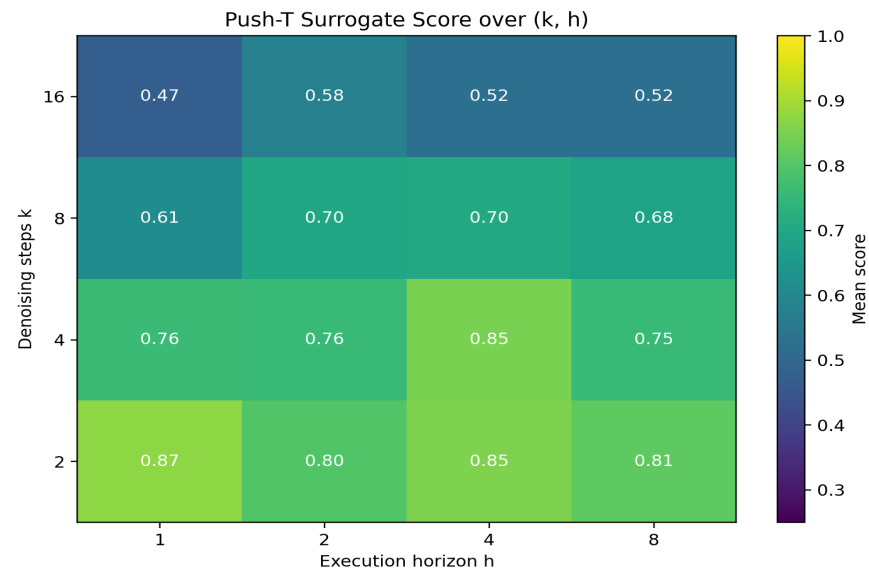


LeRobot PushT Checkpoint

- Model card 500-episode max reward: 0.955
- Model card success rate: 65.4%
- Local safetensors loads and runs in real gym-pusht
- Local smoke: max reward 0.627, 100 denoising steps
- Caveat: mirror file did not pass HF LFS SHA check

Static Grid

- Grid: k in $\{2,4,8,16\}$, h in $\{1,2,4,8\}$
- Contact-rich pushing makes equal-compute points behave differently
- The best fixed $B=2$ score point is $(2,1)$



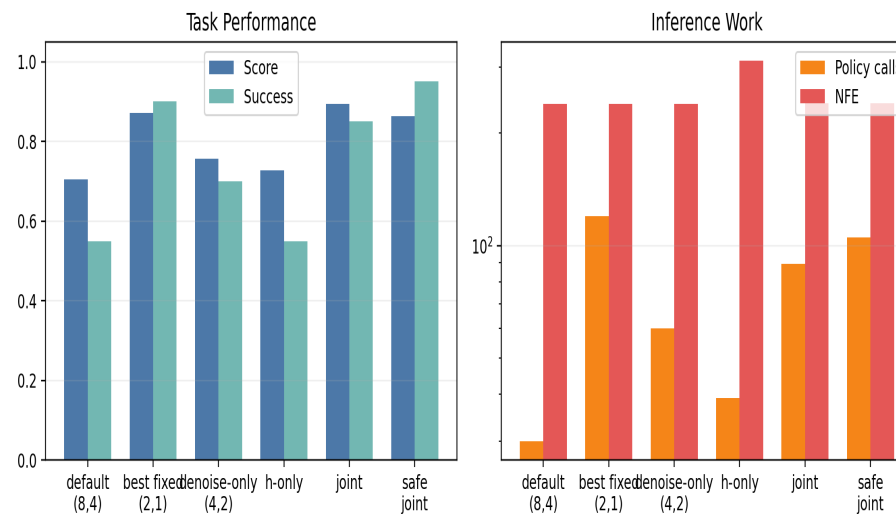
Joint Scheduler

- Training-free rule on the $B=2$ frontier
- Score variant uses (4,2) in easier phases and (2,1) near contact
- Safe variant switches to (2,1) earlier
- Goal: keep compute fixed while moving reactivity to high-risk phases

Main Results

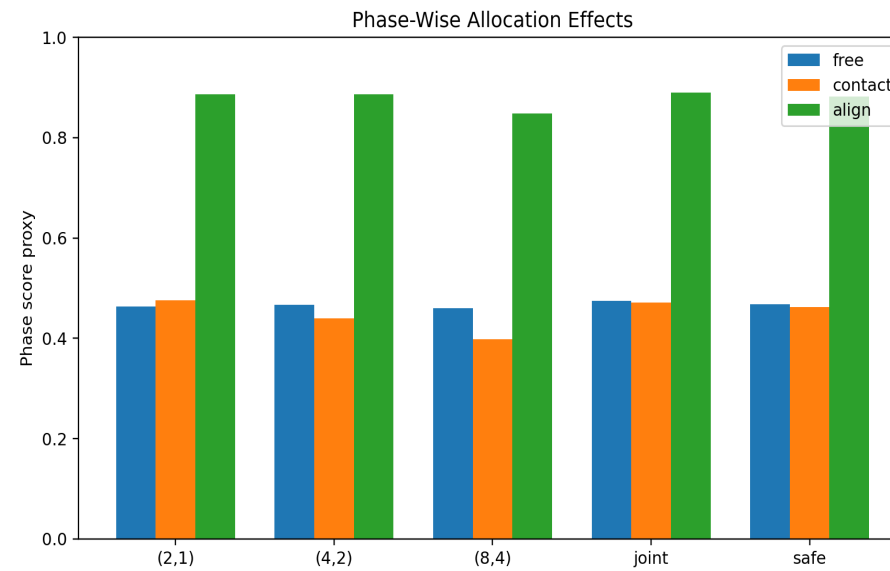
- Score scheduler: score 0.893, calls 89.2
- Safe scheduler: success 0.950, calls 105.3
- Best fixed B=2: score 0.871, success 0.900
- Result is a Pareto tradeoff, not a single universal winner

Main Baseline Comparison under $B \approx 2$



Phase-Wise Analysis

- Free-space approach tolerates longer chunks
- Contact pushing needs frequent replanning
- Near-goal alignment benefits from smoother denoised chunks



Limitations

- Improvement numbers are from the surrogate, not official benchmark rollouts
- Official run uses compatibility patches and scores below checkpoint filename
- Official fixed-(k,h) frontier favors high denoising
- LeRobot checkpoint path is executable but not SHA-verified
- $B=k/h$ is an approximate compute model
- No real-robot validation

Takeaways

- Denoising and replanning are coupled deployment decisions
- Best allocation changes by task phase and checkpoint
- A simple joint scheduler can improve the score-smoothness or success-cost tradeoff under the same budget